Generalized linear models

Logistic regression with categorical covariates

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Winter term 20/21

Logistic regression with categorical covariates - Contents

Binary covariates

- x binary, y binary $\rightarrow 2 \times 2$ contingency table
- Example:
  x gender, y duration of unemployment (short vs. long)

			unemployment short 1	long 0	Σ
gender	1	m	403	167	570
	2	f	238	175	413
			641	342	983

$$OR = \frac{403/167}{238/175} = \frac{403 \cdot 175}{167 \cdot 238} \approx 1{,}77$$

Binary covariates

How to measure this relation?

- Test: χ^2-Test, Fisher-Test
- Odds ratio (proportion of chances)
- empirical chances for short term unemployment

$$♂: \hat{\gamma}(g=1) = \frac{403}{167} = 2.413$$

$$♀: \hat{\gamma}(g=2) = \frac{238}{175} = 1.36$$

$$\text{Odds ratio}: \hat{\gamma}(1|2) = \frac{\hat{\gamma}(g=1)}{\hat{\gamma}(g=2)} = \frac{2.413}{1.36} = 1.77$$

Binary covariates

True underlying structure

		y	
		1	0
g	1	$\pi(g=1)$	$1-\pi(g=1)$
	2	$\pi(g=2)$	$1-\pi(g=2)$

- Odds: $\gamma(g=i) = \dfrac{\pi(g=i)}{1-\pi(g=i)}$
- Odds ratio: $\gamma(g=1 \mid g=2) = \dfrac{\gamma(g=1)}{\gamma(g=2)} = \dfrac{\dfrac{\pi(g=1)}{1-\pi(g=1)}}{\dfrac{\pi(g=2)}{1-\pi(g=2)}}$

Logit model for binary x

$$x_g = \begin{cases} 1, & g = 1 \ (♂) \\ 0, & g = 2 \ (♀) \end{cases}$$

- Encoded covariate:
- Model: $\boxed{\log\left(\frac{\pi(g)}{1-\pi(g)}\right) = \beta_0 + \beta_1 x_g}$
- Interpretation β_0 :

$$\Rightarrow \beta_0 = \log\left(\frac{\pi(g=0)}{1-\pi(g=0)}\right)$$ log-odds in reference category

$$\Rightarrow \exp(\beta_0) \Rightarrow$$ chance/odds in reference category

Logit model for binary x

- Interpretation β_1 :

$$\beta_1 = \log\left(\frac{\pi(g=1)}{1-\pi(g=1)}\right) - \log\left(\frac{\pi(g=0)}{1-\pi(g=0)}\right)$$

$$= \log(\gamma(x_g=1)) - \log(\gamma(x_g=0))$$

$$= \log\left(\frac{\gamma(x_g=1)}{\gamma(x_g=0)}\right) \rightarrow \text{log-odds ratio}$$

$$\Rightarrow \exp(\beta_1) = \frac{\gamma(x_g=1)}{\gamma(x_g=0)} \rightarrow \text{odds ratio}$$

$\Rightarrow$ logit model parametrizes the odds ratio

Logit model for binary *x*

Example:

$$\hat{\beta}_0 = 0.307 \qquad \hat{\beta}_1 = 0.570$$ hard to interpret

$$\underbrace{\exp(\hat{\beta}_0) = 1.36}_{\text{chances/odds female (reference)}} \qquad \underbrace{\exp(\hat{\beta}_1) = 1.77}_{\text{odds ratio men/women (chances ratio)}}$$ easier to interpret

In general:

$\hat{\beta}_1 = 0 \Rightarrow$ no relationship $\Rightarrow \exp(0) = 1$

$\hat{\beta}_1 > 0 \Rightarrow OR > 1$ (advantage males)

$\hat{\beta} < 0 \Rightarrow OR < 1$ (– " – female)

Logit model for binary *x*

Study: Heart diseases of South African men

- Response variable: congestive heart disease ($\overset{1}{\text{yes}}$/$\overset{0}{\text{no}}$)
- examined influential variables (selection):
  - Tobacco consumption ($\overset{1}{\text{yes}}$/$\overset{0}{\text{no}}$)
  - Alcohol consumption ($\overset{1}{\text{yes}}$/$\overset{0}{\text{no}}$)
  - Pre-existing illness in the family ($\overset{1}{\text{yes}}$/$\overset{0}{\text{no}}$)

Logit model for binary *x*

Study: Heart diseases of South African men

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	-1.687767	0.187832	-8.986	< 2e-16	***
tobacco	0.140233	0.025059	5.596	2.19e-08	***
alcohol	-0.001133	0.004291	-0.264	0.792	
famhistPresent	1.150282	0.212859	5.404	6.52e-08	***

Interpretation?

Smoker (with pre-existing disease in family) compared to
non-smoker (— " —)

$$OR = \frac{\exp(\cancel{-1.69} + 0.14 \cdot 1 + \cancel{1.15 \cdot 1})}{\exp(\cancel{-1.69} + 0.14 \cdot 0 + \cancel{1.15 \cdot 1})} = \exp(0.14) = 1.15$$

Multi-categorical covariates

Example:

Level of education $\Longrightarrow$ duration of unemployment

$x \backslash y$	short	long	odds
1	202	96	$202/96 = 2.1 = \hat{\gamma}(1)$
2	307	162	$307/162 = 1.9 = \hat{\gamma}(2)$
3	87	66	$87/66 = 1.3 = \hat{\gamma}(3)$
4	45	18	$45/18 = 2.5 \quad \hat{\gamma}(4)$

L

Multi-categorical covariates

$L = \{1, \dots, k\}$

Logit model:

Encoding: $L \in \{1, \dots, k-1\} \Rightarrow x_L(j) = \begin{cases} 1 & L = j \\ 0 & \text{else} \end{cases}$ for $j = 1, \dots, k-1$

$$\log \frac{\pi(L)}{1-\pi(L)} = \beta_0 + \beta_{L(1)} x_{L(1)} + \dots + \beta_{L(k-1)} x_{L(k-1)}$$

Interpretation β_0:

$\beta_0 = \log\left(\frac{\pi(k)}{1-\pi(k)}\right)$ log-odds in reference

$\Rightarrow \exp(\beta_0)$ - odds in reference $(\pi(k))$

Multi-categorical covariates

Interpretation of other coefficients:

$$\beta_{L(j)} = \log(\gamma(j)) - \log(\gamma(h)) = \log\left(\frac{\gamma(j)}{\gamma(h)}\right)$$

$\Rightarrow$ log-OR compared to reference

$$\Rightarrow \exp(\beta_{L(j)}) = \frac{\gamma(j)}{\gamma(h)} \hat{=} OR$$

For instance:

	$\beta_{L(j)}$	$\exp(\beta_{L(j)})$
1	-0.17	0.84
2	-0.27	0.76
3	-0.65	0.52
4	0	1

$\Rightarrow$ simple interpretation is possible

Models with interactions

Example: y = approval to the 3-month abortion law

covariates				Response	
gender (G)		age (A)		yes	no
m	0	young	1	0.6	0.4
		old	0	0.5	0.5
		young	1	0.8	0.2
f	1	old	0	0.5	0.5

$$\gamma_{young}(m \mid f) = \frac{0.6/0.4}{0.8/0.2} = 3/8 \neq 1$$

$$\gamma_{old}(m \mid f) = \frac{0.5/0.5}{0.5/0.5} = 1$$

Models with interactions

Logit model for this example:

$$\log\left(\frac{P(\text{yes} \mid G, A)}{P(\text{no} \mid G, A)}\right) = \beta_0 + \beta_g \overbrace{x_g}^{0,1} + \beta_A \overbrace{x_A}^{0,1}$$

$$\Rightarrow \frac{P(\text{yes} \mid G, A)}{P(\text{no} \mid G, A)} = \exp(\beta_0) \cdot \exp(\beta_g x_g) \cdot \exp(\beta_A x_A)$$

$$\Rightarrow \gamma_A(m \mid f) = \frac{\dfrac{P(\text{yes} \mid G=m, A)}{P(\text{no} \mid G=m, A)}}{\dfrac{P(\text{yes} \mid G=f, A)}{P(\text{no} \mid G=f, A)}} = \frac{\cancel{\exp(\beta_0)} \cdot \exp(\beta_g) \cancel{\exp(\beta_A x_A)}}{\cancel{\exp(\beta_0)} \cancel{\exp(\beta_A x_A)}}$$

$$= \exp(\beta_g)$$

A is (arbitrarily) fixed

Models with interactions

Issue:

- Model can't adequately display the relations from above.
- Better: Use model with interactions!

Model:

$$\log\left(\frac{P(yes \mid G, A)}{P(no \mid G, A)}\right) = \beta_0 + \beta_G x_G + \beta_A x_A + \beta_{GA} \underbrace{x_A x_G}_{=: x_{AG}}$$

$$\Rightarrow \frac{P(yes \mid G, A)}{P(no \mid G, A)} = \exp(\beta_0) \cdot \exp(\beta_G x_G) \cdot \exp(\beta_A \cdot x_A) \cdot \exp(\beta_{GA} \cdot x_A \cdot x_G)$$

Models with interactions

Interpretation of the coefficients:

- $\exp(\beta_0)$ odds in the reference category
- $\exp(\beta_G)$ change by $x_G = 1$ keeping x_A constant
- $\exp(\beta_A)$ — " — $x_A = 1$ — " — x_G — " —
- $\exp(\beta_{GA})$ additional change if $x_G = 1$ and $x_A = 1$

$\Rightarrow$ difficult to be interpreted alone

Overview

But if $\exp(\beta_{GA}) = 1 \iff \beta_{GA} = 0$

$\iff$ relationship age $\longleftrightarrow$ response is not gender-specific

gender $\longleftrightarrow$ response is not age-spezific